Course Summary and Looking Forward

Convex Opt :

LP

SDP

Conv. Opt.

Duality

Grad Desc.

Sub grad method

Prox Grad

Newton Method & Q-N Method

Stochastic Methods

Looking Forward: Modeling + Software

LP SDP Convex

Modeling: within special classes of convex opt'n.

Software:

Modeling for Non-convex. → continuous
→ Comb. opt & integer programming.

Looking Forward: Building New Algorithms

- Deeper into analysis
- Dual algorithms
- Newton → Quasi-Newton Methods.
- Randomized (SGD)
- IPM

Next Directions

Modeling

Analysis

Algorithm

Applications in ML.

Dynamic Problems

Online opt.

Bandit Problems

Reinforcement Learning